

Supplementary information

Reproductive life-course phenotyping in the National Survey of Family Growth: a self-supervised temporal survey study

Feifan Lu, Fengshang Yan, Qianqian Yang, Jiuqiong Yan, Rui Guan

Supplementary tables

Supplementary Table S1 through Supplementary Table S11 provide numerical source summaries for cluster selection, endpoint enrichment intervals, age-range sensitivity, covariate adjustment, baseline methods, leakage sensitivity, subgroup robustness, the ever-pregnant stratum, simple age/parity baselines, raw-feature versus SSL supervised comparisons, and encoder seed sensitivity.

Supplementary Table S1: Development-cycle cluster selection metrics across k=2–8.

k	Silhouette	DBI	Min. cluster, %	Boot. ARI	Boot. n	Selected
2	0.587	0.617	43.3	0.999	60	False
3	0.606	0.701	4.6	0.914	60	True
4	0.518	0.720	4.7	0.738	60	False
5	0.385	0.969	4.0	0.797	60	False
6	0.385	0.953	1.6	0.780	60	False
7	0.366	1.040	3.2	0.750	60	False
8	0.366	1.029	1.5	0.715	60	False

Supplementary Table S2: Full temporal-validation phenotype-by-endpoint enrichment with bootstrap intervals. Intervals are stratified cluster percentile bootstrap intervals using public-use VEST/VECL design variables with respondent weights retained.

Endpoint	Phenotype	Events	Prev., %	PR (95% CI)	RD, % (95% CI)
Adverse proxy	P0	7	0.5	0.09 (0.01–0.19)	-4.9 (-5.7 to -4.1)
Adverse proxy	P1	219	9.2	1.72 (1.58–1.85)	3.8 (3.0 to 4.7)
Adverse proxy	P2	48	22.2	4.13 (2.98–5.46)	16.8 (10.7 to 24.0)
Contraceptive vuln.	P0	146	5.6	0.72 (0.63–0.81)	-2.2 (-2.9 to -1.5)
Contraceptive vuln.	P1	238	10.4	1.33 (1.22–1.44)	2.6 (1.7 to 3.5)
Contraceptive vuln.	P2	21	7.6	0.97 (0.59–1.39)	-0.2 (-3.2 to 3.0)
Fertility/loss help	P0	114	4.3	0.39 (0.33–0.45)	-6.8 (-7.7 to -5.9)
Fertility/loss help	P1	397	18.2	1.64 (1.55–1.75)	7.1 (6.1 to 8.4)
Fertility/loss help	P2	44	18.6	1.67 (1.19–2.19)	7.5 (2.1 to 13.3)
Fecundity limit./infert.	P0	466	17.8	0.63 (0.58–0.68)	-10.3 (-11.9 to -8.7)
Fecundity limit./infert.	P1	807	38.8	1.38 (1.32–1.44)	10.6 (9.0 to 12.3)
Fecundity limit./infert.	P2	94	41.0	1.46 (1.22–1.69)	12.9 (6.2 to 19.3)
Mistimed/unwtd.	P0	51	2.3	0.07 (0.05–0.09)	-31.9 (-34.1 to -29.6)
Mistimed/unwtd.	P1	1,506	67.0	1.96 (1.88–2.05)	32.8 (31.2 to 34.4)
Mistimed/unwtd.	P2	181	76.4	2.23 (2.06–2.43)	42.2 (36.8 to 47.6)

Supplementary Table S3: Age-range sensitivity analysis in the 2022–2023 temporal-validation cycle. The phenotype with the highest 15–44 prevalence ratio for each endpoint is shown, then re-evaluated after including females aged 45–49 years. Phenotype assignment used the primary endpoint-excluded encoder without refitting; full-age assignment labels were aligned to the primary P0/P1/P2 labels by majority overlap in the original 15–44 subset, and reassignment agreement was checked only as a technical reproducibility control.

Endpoint	Phenotype	15–44 N	15–44 PR	15–49 N	15–49 PR	Direction
Contraceptive vulnerability	P1	2,228	1.33	2,691	1.26	Same direction
Fertility or loss help	P2	239	1.68	315	1.35	Same direction
Mistimed/unwanted pregnancy history	P2	239	2.23	315	1.98	Same direction
Adverse pregnancy-history proxy	P2	239	4.11	315	3.25	Same direction
Fecundity limitation/infertility	P2	239	1.47	315	1.34	Same direction

Supplementary Table S4: Covariate-adjusted phenotype endpoint enrichment in 2022–2023. Models are survey-weighted one-vs-rest logistic enrichment models adjusted for age, race/ethnicity, education, poverty, insurance, and parity. Confidence intervals are stratified cluster percentile bootstrap intervals using public-use VEST/VECL. These models are descriptive robustness checks and are not causal models.

Endpoint	Phenotype	Adjusted OR	95% CI	Bootstrap n
Adverse pregnancy-history proxy	P0	0.11	0.01–0.29	300
Adverse pregnancy-history proxy	P1	1.65	1.04–2.85	300
Adverse pregnancy-history proxy	P2	2.05	1.20–3.57	300
Contraceptive vulnerability	P0	0.52	0.38–0.70	300
Contraceptive vulnerability	P1	1.84	1.39–2.41	300
Contraceptive vulnerability	P2	0.69	0.40–1.03	300
Fertility or loss help	P0	0.25	0.17–0.33	300
Fertility or loss help	P1	2.50	1.92–3.56	300
Fertility or loss help	P2	1.23	0.78–1.86	300
Fecundity limitation/infertility	P0	0.74	0.60–0.89	300
Fecundity limitation/infertility	P1	1.25	1.05–1.52	300
Fecundity limitation/infertility	P2	0.99	0.67–1.33	300
Mistimed/unwanted pregnancy history	P0	0.02	0.01–0.03	300
Mistimed/unwanted pregnancy history	P1	9.58	7.69–12.81	300
Mistimed/unwanted pregnancy history	P2	2.25	1.56–3.47	300

Supplementary Table S5: Baseline phenotyping method comparison in the 2022–2023 temporal-validation cycle. Mean maximum prevalence ratio is averaged across prespecified endpoints. Minimum cluster proportion and bootstrap ARI are development/test stability summaries; a high prevalence ratio from a very small cluster should not be interpreted as a stronger method.

Method	Mean max PR	Boot. ARI	Min. cluster, %	Endpoints
MCA-style one-hot SVD + k-means	1.57	0.774	27.5	5
Raw matrix PCA + k-means	2.25	0.735	0.1	5
SSL embedding + k-means	2.16	0.953	4.9	5
Selected-variable Bernoulli LCA	1.50	0.904	18.5	5

Supplementary Table S6: Endpoint-excluded versus full-domain encoder sensitivity. Values show the maximum phenotype prevalence ratio for each endpoint in 2022–2023. The primary endpoint-excluded encoder retained endpoint-enrichment patterns despite direct endpoint-variable exclusion.

Endpoint	Endpoint-excluded	Full-domain
Contraceptive vulnerability	1.33	1.29
Fertility or loss help	1.67	1.59
Mistimed/unwanted pregnancy history	2.23	2.00
Adverse pregnancy-history proxy	4.13	2.01
Fecundity limitation/infertility	1.46	1.39

Supplementary Table S7: Subgroup robustness summary. The table lists the highest subgroup-level phenotype endpoint enrichments with at least 50 respondents in the phenotype-subgroup cell. Full subgroup results are provided in source-data tables.

Subgroup	Endpoint	Phenotype	N	Events	PR
Parity: Parity 0	Mistimed/unwtd.	P1	227	186	10.68
Parity: Parity 0	Adverse proxy	P1	227	11	9.90
Age: 15-24	Adverse proxy	P1	155	16	8.23
Age: 15-24	Mistimed/unwtd.	P1	155	124	7.82
Race/ethnicity: Non-Hispanic White	Adverse proxy	P2	84	14	6.67
Insurance: Private/Medi-Gap	Adverse proxy	P2	101	16	6.06
Age: 15-24	Fertility/loss help	P1	155	18	5.37
Poverty: $\geq 200\%$ FPL	Adverse proxy	P2	96	11	4.02
Parity: Parity 0	Fertility/loss help	P1	227	51	3.99
Age: 25-34	Adverse proxy	P2	83	14	3.59
Poverty: 100-199% FPL	Adverse proxy	P2	54	10	3.39
Poverty: $<100\%$ FPL	Adverse proxy	P2	92	27	2.92

Supplementary Table S8: Ever-pregnant stratum analysis for pregnancy-history endpoints in the 2022–2023 temporal-validation cycle. This analysis restricts to respondents with at least one pregnancy record in the public-use pregnancy file to reduce mechanical enrichment from pregnancy exposure itself. Intervals are stratified cluster bootstrap percentile intervals using public-use VEST/VECL.

Endpoint	Phenotype	Events/N	Prev., %	PR (95% CI)	RD, % (95% CI)
Adverse proxy	P0	7/73	15.9	1.48 (0.24–2.81)	5.2 (-7.9 to 20.4)
Adverse proxy	P1	219/2,223	9.2	0.86 (0.79–0.92)	-1.5 (-2.4 to -0.8)
Adverse proxy	P2	48/229	23.2	2.16 (1.60–2.85)	12.5 (6.4 to 19.6)
Mistimed/unwtd.	P0	51/73	73.2	1.07 (0.90–1.22)	4.8 (-6.6 to 15.3)
Mistimed/unwtd.	P1	1,506/2,223	67.0	0.98 (0.97–0.99)	-1.4 (-2.2 to -0.6)
Mistimed/unwtd.	P2	181/229	80.0	1.17 (1.09–1.25)	11.6 (6.0 to 16.8)

Supplementary Table S9: Reviewer-requested simple stratification baselines for phenotype endpoint enrichment. Age-by-parity and age-by-ever-pregnant strata were compared with the SSL phenotype to test whether endpoint enrichment was largely recoverable from simple reproductive-exposure strata. These are descriptive enrichment summaries, not causal adjustments.

Analysis	Endpoint	SSL phenotype	Age x parity	Age x ever-pregnant
Full cohort	Contraceptive vuln.	1.33 (P1)	1.99 (15-24 / parity 1-2)	1.53 (25-34 / ever pregnant)
Full cohort	Fertility/loss help	1.67 (P2)	1.93 (35-44 / parity 1-2)	1.95 (35-44 / ever pregnant)
Full cohort	Mistimed/unwtd.	2.23 (P2)	2.39 (35-44 / parity 3+)	2.40 (15-24 / ever pregnant)
Full cohort	Adverse proxy	4.13 (P2)	2.91 (35-44 / parity 3+)	2.15 (35-44 / ever pregnant)
Full cohort	Fecundity limit./infert.	1.46 (P2)	1.79 (35-44 / parity 3+)	1.71 (35-44 / ever pregnant)
Ever-pregnant	Mistimed/unwtd.	1.17 (P2)	1.21 (15-24 / parity 0)	–
Ever-pregnant	Adverse proxy	2.16 (P2)	1.46 (35-44 / parity 3+)	–

Supplementary Table S10: Reviewer-requested supervised comparison of simple reproductive-exposure variables, raw primary encoder inputs, SSL embeddings, and combined feature sets. Models were L2-regularized logistic models trained on 2011–2019 and evaluated once in 2022–2023. The upper matrix reports AUPRC, the lower matrix reports AUROC. These unweighted discrimination summaries test representation utility and are not clinical prediction models. Full AUPRC-enrichment and delta values are provided in the source-data tables.

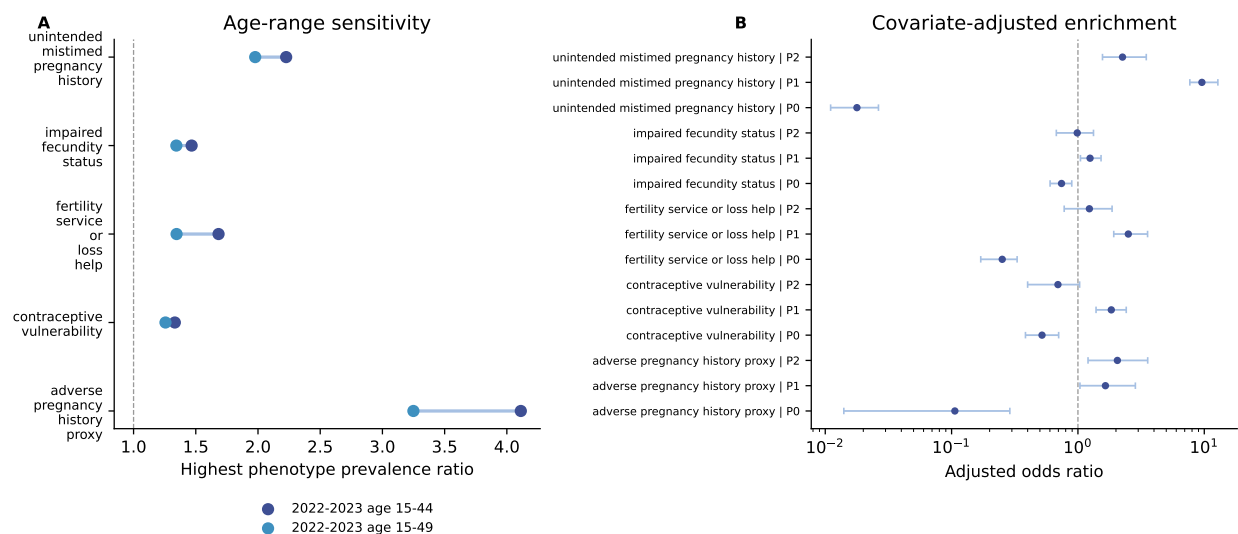
<i>AUPRC matrix</i>										
Analysis	Endpoint	Train ev.	Test ev.	Base, %	Age/parity	Raw 48	SSL	Raw+SSL	Pheno.	SSL+ph.
Ever-pregnant	Mistimed/unwtd.	10,447	1,738	68.8	0.766	0.834	0.846	0.870	0.699	0.845
Ever-pregnant	Adverse proxy	2,202	274	10.9	0.158	0.173	0.190	0.183	0.127	0.192
Full cohort	Contraceptive vuln.	1,707	405	8.3	0.094	0.124	0.472	0.490	0.096	0.457
Full cohort	Fertility/loss help	2,194	555	11.3	0.207	0.234	0.276	0.284	0.165	0.277
Full cohort	Mistimed/unwtd.	10,447	1,738	35.5	0.766	0.790	0.831	0.862	0.681	0.833
Full cohort	Adverse proxy	2,202	274	5.6	0.167	0.186	0.190	0.184	0.123	0.191
Full cohort	Fecundity limit./infert.	5,361	1,367	27.9	0.429	0.455	0.529	0.536	0.338	0.527
<i>AUROC matrix</i>										
Analysis	Endpoint	Age/parity		Raw 48	SSL	Raw+SSL	Pheno.	SSL+ph.		
Ever-pregnant	Mistimed/unwtd.	0.60		0.71	0.72	0.77	0.52	0.72		
Ever-pregnant	Adverse proxy	0.59		0.64	0.67	0.67	0.55	0.67		
Full cohort	Contraceptive vuln.	0.55		0.65	0.90	0.91	0.57	0.90		
Full cohort	Fertility/loss help	0.73		0.75	0.78	0.79	0.66	0.78		
Full cohort	Mistimed/unwtd.	0.90		0.91	0.92	0.94	0.87	0.92		
Full cohort	Adverse proxy	0.81		0.82	0.84	0.84	0.77	0.84		
Full cohort	Fecundity limit./infert.	0.69		0.70	0.74	0.75	0.61	0.74		

Supplementary Table S11: SSL encoder initialization sensitivity. The primary development-selected k=3 solution was held fixed and a complete 30-epoch alternative initialization was compared with the primary seed. This analysis tests whether the learned representation and transferred phenotype assignment are sensitive to encoder initialization; it is not a repeated full model-selection study.

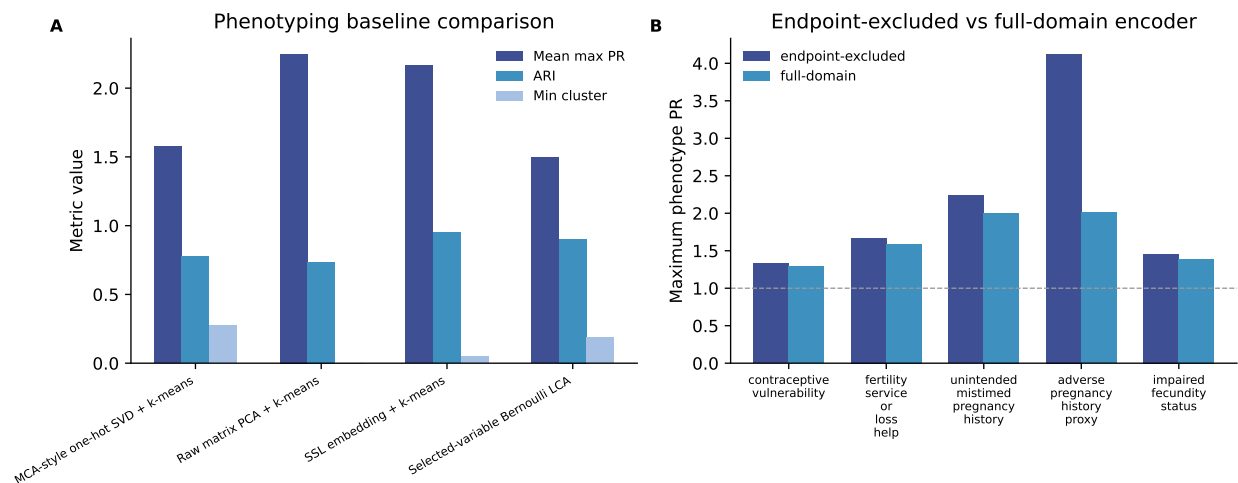
Seed	Epochs	k	Masked MSE	Silhouette	DBI	Test ARI	Adverse PR
20,260,602	30	3	0.917	0.615	0.70	1.00	2.16
20,260,612	30	3	0.724	0.453	0.94	0.43	1.68

Supplementary figures

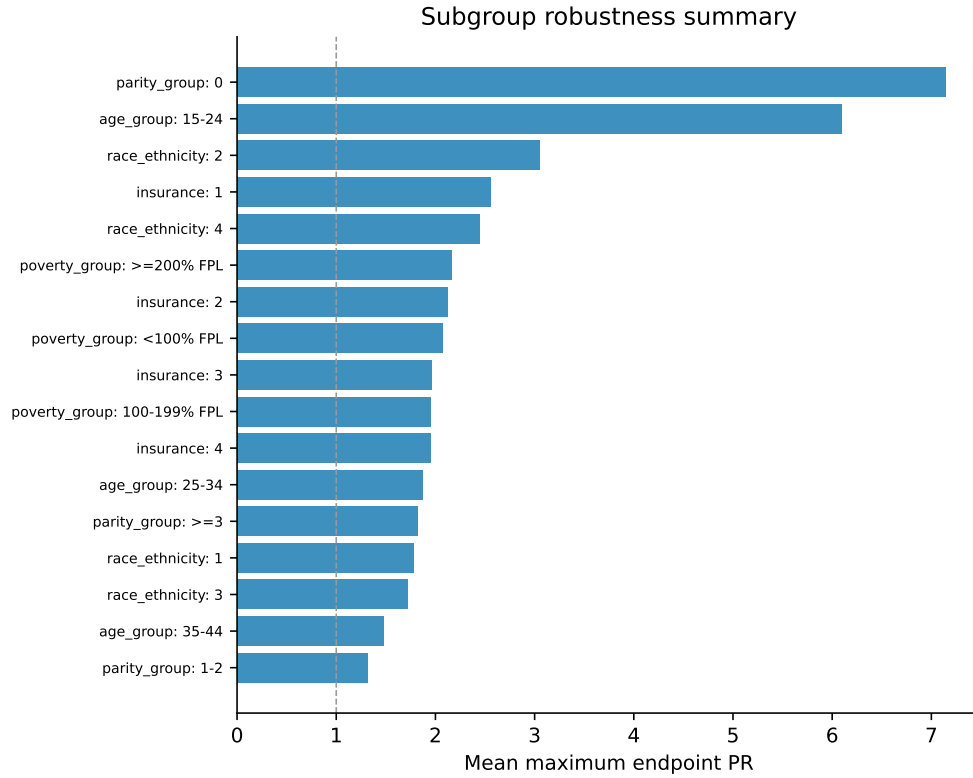
Supplementary Figures S1–S3 provide robustness, method-sensitivity, and subgroup displays; Supplementary Figures S4 and S5 provide the secondary supervised validation and SSL diagnostic displays.



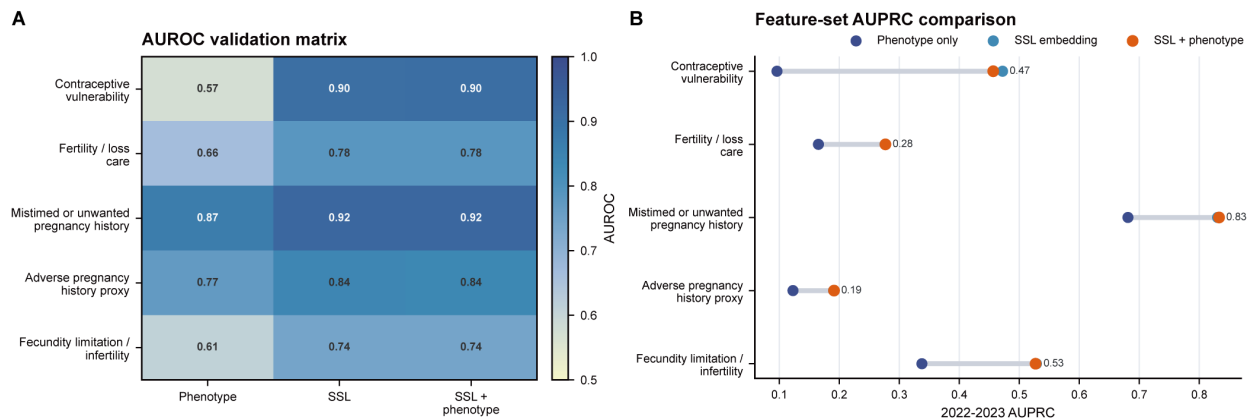
Supplementary Figure S1: Robustness analysis for age-range sensitivity and covariate-adjusted endpoint enrichment. (A) Highest-phenotype prevalence ratios in 2022–2023 ages 15–44 versus 15–49 years. (B) Survey-weighted adjusted odds ratios from one-vs-rest phenotype enrichment models.



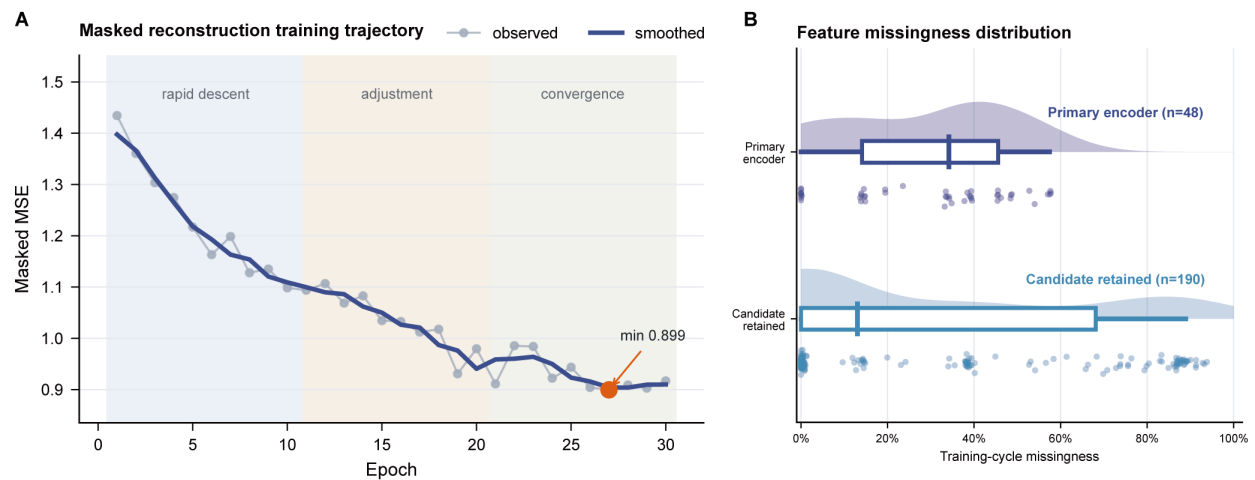
Supplementary Figure S2: Baseline-method and leakage-sensitivity analyses. (A) SSL phenotyping compared with raw PCA, MCA-style one-hot SVD, and selected-variable Bernoulli LCA baselines. (B) Maximum endpoint prevalence ratios for the primary endpoint-excluded encoder and a full-domain sensitivity encoder.



Supplementary Figure S3: Subgroup robustness summary across age, race/ethnicity, poverty, insurance, and parity strata. Values summarize mean maximum endpoint prevalence ratios by subgroup; subgroup categories use public-use recodes, and full decoded endpoint-by-phenotype estimates are included in source data and Supplementary Table S7.



Supplementary Figure S4: Secondary supervised validation diagnostics. (A) AUROC validation matrix for phenotype-only, SSL-embedding, and SSL-plus-phenotype summaries across validation endpoints. AUROC is reported as a secondary discrimination summary and is not the primary claim. (B) Feature-set AUPRC comparison for phenotype-only, SSL-embedding, and SSL-plus-phenotype summaries. These diagnostics support representation and enrichment assessment rather than diagnostic prediction.



Supplementary Figure S5: SSL optimization and feature-missingness diagnostics. (A) Masked reconstruction loss across SSL training epochs, with rapid-descent, adjustment, and convergence phases, smoothed trend, and minimum-loss epoch annotation. (B) Distribution of training-cycle missingness among primary encoder features and retained candidate SSL features.